# Md Erfanul Islam Bhuiyan

Backend & AI Engineer

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## Profile

Backend and AI-focused Software Engineer with **around 5 years** building enterprise SaaS platforms and event-driven distributed systems. I turn complex business requirements into reliable services in **Python and .NET**, build RAG and AI-agent workflows, and use AI across the software development lifecycle to speed up delivery.

## Professional Experience

**Senior Backend Developer** Aug 2026 – Oct 2026
*InfinitiBit* *Dhaka, Bangladesh*

- Architected the **Python/FastAPI and PostgreSQL** backend of an AI VAT-validation product that checks invoices against German and EU tax law and cites the legal provision behind every verdict.
- Evolved the proof of concept into a multi-tenant SaaS MVP with SSO, role-based access, credit billing and audit logs, isolating each customer's data by PostgreSQL schema and database role on every query.
- Delivered tooling to create, monitor and sync each customer's tax-law knowledge graph, with versioning for easy rollback and human approval before any law change goes live.
- Defined a RabbitMQ event contract for the company's SaaS products and engineered Temporal workflows with a transactional outbox, sized to run **10,000** validations from one click and recover after restarts.
- Instrumented the API, workers and LLM calls end to end with OpenTelemetry and Grafana; set up Bifrost with DevOps as an AI gateway with per-app virtual keys to control token costs across LLM providers.
- Introduced an AI-assisted delivery workflow with Claude Code and reusable agent skills; reviewed **35+** pull requests and mentored two junior developers.

**Senior Software Engineer (Backend)** Apr 2026 – Jul 2026
*SDS Manager (Netpower)* *Dhaka, Bangladesh*

- Created a .NET 10 document-intelligence service with smart chunking, sparse vectors and dense embeddings using Microsoft.Extensions.AI.
- Led development of a shared background-job system for quality-management microservices, with custom recurrence scheduling and OpenTelemetry tracing.
- Packaged RabbitMQ messaging into a reusable library for notification and email events between microservices, and set up CI/CD pipelines for multiple service repositories.

**Senior Backend Developer** Nov 2025 – Mar 2026
*InfinitiBit* *Dhaka, Bangladesh*

- Led backend development of a private GraphRAG chat system on a local GPU cluster, cutting response time from **10–12 minutes to under 2 minutes** with an asynchronous agent pipeline.
- Engineered prompts and context with cumulative history summarization for long conversations, and evaluated answer quality using LLM-as-a-judge.
- Created a document ingestion tool that converted **3 TB** of PDFs, Office files and images into text with metadata for the knowledge graph and model training.
- Developed a VAT audit agent with GraphBit in a 3-day hackathon that extracts rules from legal text and explains each finding.
- Architected the backend of a transfer-pricing tax platform, and delivered its single sign-on, role-based access, document storage and working-paper ingestion.
- Reviewed **45+** pull requests across the chat and tax platforms and planned work for two developers.

**Backend Developer (.NET)** Nov 2024 – Nov 2025
*InfinitiBit* *Dhaka, Bangladesh*

- Orchestrated AI analysis of bid documents (RFQ, technical, pricing) with background jobs, progress tracking and a circuit breaker.
- Built core modules of a .NET 8 bid-management platform for a leading German energy company as its largest contributor, with **150+** merged pull requests.
- Automated bid approval and e-signature with workflow rules, document version locking and approval reports; connected modules over RabbitMQ and integrated AWS and SharePoint.
- Designed the architecture and AWS cost model of a power-utility platform sized for **7,500** sensor messages/sec.

**Software Engineer** Jan 2022 – Jul 2024
*Enosis Solutions* *Dhaka, Bangladesh*

- Fixed unauthorised data access in production with a central policy-based access control (PBAC) layer across APIs.
- Implemented Microsoft Entra SSO (OAuth2/OIDC) for a multi-tenant B2B SaaS platform with **100k+** users.
- Built a serverless, independently scaling email service that sends mail from clients' own accounts and templates.
- Migrated a hiring platform from Node.js to .NET 8 as a modular monolith, and a US dental practice system from VB.NET 3.5 to C# .NET 6.

## Technical Skills

**Languages** — Python, C#, TypeScript, SQL, Go

**AI & LLM** — Agent Workflows, RAG, GraphRAG, Knowledge Graphs, LangGraph, MCP Servers & Tool Calling, Vector Search (Qdrant, pgvector), Local LLMs, CrewAI, LLM Gateways (LiteLLM, Bifrost)

**Backend & Architecture** — FastAPI, REST APIs, ASP.NET Core, Microservices, Event-Driven Systems, Domain-Driven Design, RabbitMQ, Temporal

**Data & Cloud** — PostgreSQL, SQL Server, Neo4j, Redis, AWS, Azure

**DevOps & Observability** — Docker, GitHub/GitLab CI/CD, OpenTelemetry, Grafana (Prometheus, Tempo, Loki)

**AI-Assisted Development** — Claude Code, Agent Skills, AI-Driven SDLC

## Personal Projects

**Paceframe**, *Goal and performance management platform* paceframe.erfanul.com

- Launched a web app for small organisations with goal tracking, check-ins and weighted quarterly reviews.
- Designed the domain model and system architecture, and implemented the modular-monolith **.NET** backend, React frontend and an **MCP server** for AI agents.

*Technologies:* C# .NET, PostgreSQL, React, TypeScript, MCP Server, Docker

**ExpenseXpert**, *Personal finance and bill-splitting app* expensexpert.erfanul.com

- Shipped an offline expense tracker and bill splitter for **Android and iOS** in **Kotlin Multiplatform**, settling group trips across a shared pot, direct payments and personal wallets to the cent.
- Owned product and architecture: a pure-Kotlin money domain built test-first with Claude Code agents, **650+** tests per commit, and load-tested with **50,000**-transaction groups.

*Technologies:* Kotlin Multiplatform, Compose Multiplatform, Room, Koin, SQLite

**CPorBit**, *Personalised competitive programming mentor* github.com/smhimran/CPorBit

- Developed a **Python/Django REST** and React web app that analyses users' Codeforces submissions and suggests practice problems to raise their contest ratings.
- Drove the system design and R&D, and authored the **KNN**-based machine-learning problem recommender.

*Technologies:* Django REST, Python, PostgreSQL, React, TailwindCSS

## Achievements

**Employee of the Month**, *InfinitiBit* Jul 2025
Awarded for contributions to backend and AI systems.

**Competitive Programming**

- Peak rank **Expert** on Codeforces (rating 1623): codeforces.com/profile/erfanul007.
- Solved **2500+** problems across online judges: linktr.ee/erfanul007.
- **Champion**, DIU Intra-University Programming Contest 2021.
- Competed in the ICPC Dhaka Regional 2019 and 2020 for Daffodil International University.

## Education

**BSc in Computer Science & Engineering** Jan 2018 – Jan 2022
*Daffodil International University* *Dhaka, Bangladesh*
CGPA: **3.73/4.00**

## Languages

**English** — C1 (IELTS Academic, Overall Band 7)

**German** — Basic (A1–A2), actively learning